

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 01-Jun-2023

Time: Three Hours

Max. Marks: 75

MEDICINAL CHEMISTRY - I

Q.P. CODE: 5014

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory

LONG ESSAYS

2 x 10 = 20 Marks

1. Define sedatives and hypnotics. Discuss the SAR of benzodiazepines. Outline the synthesis of and specific uses of Diazepam.
OR
Define and classify adrenergic agents. Discuss the SAR of adrenergic agents. Write the synthesis of salbutamol.
2. **a) What are general anaesthetics? Classify them with structural examples. Outline the synthesis of halothane and methohexitol sodium**
b) Write the SAR of morphine with respect to its nuclear modifications.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Discuss the reduction reactions in drug metabolism.
OR
Explain the role of hydrogen bonding on drug action, giving examples.
4. Explain the role of Cytochrome P-450 and mono oxygenase in biotransformation.
OR
What are adrenergic blockers? Give the structure and uses of any three of them. Write the synthesis of propranolol.
5. Discuss the biosynthesis and metabolism of acetyl choline. Give its functions.
6. Explain about reversible and irreversible cholinesterase inhibitors.
7. What are cholinergic blockers? List out their therapeutic importance. Write the synthesis of Dicyclomine hydrochloride.
8. Classify barbiturates with structural examples. Write their general mode of action and therapeutic uses.
9. Outline the synthesis and uses of Ibuprofen and Fantanyl citrate.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Distinguish between phase-I and phase-II biotransformations.
11. What do you mean bioisosterism? Give examples.
12. Write the structure and uses of Methyldopa and Pilocarpine.
13. Give the structure, uses of metoprolol and Phentolamine.
14. Write about nicotinic receptors, their types and functions.
15. Give the synthesis of Tolazoline and mention its uses.
16. Differentiate between physostigmine and neostigmine.
17. Give the structure, uses of Benzotropine mesylate and Scopolamine hydrobromide.
18. Outline the synthesis and uses of chlorpromazine hydrochloride.
19. Write the structure and specific uses of any two Narcotic antagonists.
